

The Coralia Sequence: A Unique Finite Integer Set Under Fibonacci–Lucas Terminal Constraints

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Abstract

We introduce the Coralia Sequence $\mathcal{C} = \{0, 1, 2, 3, 5, 7, 9, 12, 15, 23, 30, 35\}$, a 12-element integer set characterized by nine axioms (C1–C9). The sequence is generated by four arithmetic zones with ascending gaps followed by a descending Fibonacci–Lucas cascade. We prove that $\mathcal{C}$ is the unique 12-element completion of a 10-element seed determined by the axioms. The terminal gap triple $(8, 7, 5) = (F_6, L_4, F_5)$ is shown to be the unique strictly descending triple from $\text{Fib} \cup \text{Luc}$ summing to 20 with all elements at least 5.

1 Introduction

Let $\varphi = (1 + \sqrt{5})/2$ denote the golden ratio. This paper introduces a finite integer set that incorporates Fibonacci–Lucas structure in its terminal gap constraints.

Definition 1.1 (Fibonacci and Lucas sequences). The Fibonacci sequence $(F_n)_{n \geq 0}$ and Lucas sequence $(L_n)_{n \geq 0}$ are defined by:

$$\begin{aligned} F_0 &= 0, & F_1 &= 1, & F_n &= F_{n-1} + F_{n-2} \text{ for } n \geq 2, \\ L_0 &= 2, & L_1 &= 1, & L_n &= L_{n-1} + L_{n-2} \text{ for } n \geq 2. \end{aligned}$$

We write $\text{Fib} = \{F_n : n \geq 0\}$ and $\text{Luc} = \{L_n : n \geq 0\}$ for the sets of Fibonacci and Lucas numbers, respectively. Note that $\text{Fib} \cap \text{Luc} = \{1, 3\}$ and $4 \in \text{Luc}$ (as $L_3 = 4$).

Definition 1.2 (Ordered set and gap sequence). Let $S \subset \mathbb{Z}$ be a finite set of integers with $|S| = n$. We write $S = \{s_0 < s_1 < \dots < s_{n-1}\}$ for its strictly increasing enumeration. The *gap sequence* of S is

$$\text{gaps}(S) = (g_0, g_1, \dots, g_{n-2}) \quad \text{where} \quad g_i = s_{i+1} - s_i.$$

We refer to $g_0, \dots, g_7$ as the *first eight gaps* and $g_{n-4}, g_{n-3}, g_{n-2}$ as the *last three gaps* (when $n \geq 4$).

Lemma 1.3 (Telescoping sum). *For any finite set $S = \{s_0 < \dots < s_{n-1}\}$,*

$$\sum_{i=0}^{n-2} g_i = s_{n-1} - s_0 = \max(S) - \min(S).$$

Proof. Direct telescoping: $\sum_{i=0}^{n-2} (s_{i+1} - s_i) = s_{n-1} - s_0$. □

2 The Axiom System

Let $\mathcal{C} \subseteq \{0, 1, \dots, 35\}$ with $|\mathcal{C}| = 12$. Write $\mathcal{C} = \{c_0 < c_1 < \dots < c_{11}\}$ and $\text{gaps}(\mathcal{C}) = (g_0, \dots, g_{10})$.

Axioms C1–C9:

- C1.** (Origin) $0 \in \mathcal{C}$.
- C2.** (Cardinality containment) $|\mathcal{C}| \in \mathcal{C}$.
- C3.** (Cardinality) $|\mathcal{C}| = 12$.
- C4.** (Ceiling) $\max(\mathcal{C}) = 35$.
- C5.** (Gap-sum closure; derivable from C1, C4) $\sum_{i=0}^{10} g_i = 35$.
- C6.** (Foundation structure) The first eight gaps satisfy:
 - (a) $g_0, g_1, \dots, g_7 \in \{1, 2, 3\}$,
 - (b) $g_0 \leq g_1 \leq \dots \leq g_7$,
 - (c) (Seed specification) $\{1, 2, 3, 5, 7, 9, 15\} \subseteq \mathcal{C}$.
- C7.** (Terminal descent) $g_8 > g_9 > g_{10}$.
- C8.** (Fibonacci–Lucas terminal) $g_8, g_9, g_{10} \in \text{Fib} \cup \text{Luc}$.
- C9.** (Terminal minimum) $g_8, g_9, g_{10} \geq 5$.

Remark 2.1. Axiom C6 has three subclauses (a)–(c); we count it as one axiom, giving nine axioms total. Axioms C2 and C3 together imply $12 \in \mathcal{C}$; we state them separately to distinguish the structural principle ($|\mathcal{C}| \in \mathcal{C}$) from the instantiated cardinality ($|\mathcal{C}| = 12$). By Lemma 1.3, C5 is equivalent to $\max(\mathcal{C}) - \min(\mathcal{C}) = 35$; given C1 and C4, this is automatic. We retain C5 to emphasize the gap-sum structure.

3 The Zone Generator

Definition 3.1 (Zone generator). Define four zones Z_1, Z_2, Z_3, Z_4 as follows:

Zone	Start	Gap vector	Elements produced
Z_1	0	(1, 1, 1)	0, 1, 2, 3
Z_2	3	(2, 2, 2)	5, 7, 9
Z_3	9	(3, 3)	12, 15
Z_4	15	(8, 7, 5)	23, 30, 35

The Coralia Sequence is $\mathcal{C} = Z_1 \cup Z_2 \cup Z_3 \cup Z_4 = \{0, 1, 2, 3, 5, 7, 9, 12, 15, 23, 30, 35\}$.

Remark 3.2. The zone generator produces $\mathcal{C}$ by construction. The axioms characterize $\mathcal{C}$. We prove below that the axioms uniquely determine $\mathcal{C}$.

4 Uniqueness Theorem

Lemma 4.1 (Forced seed). *Axioms C1, C2, C3, C4, and C6(c) together force the inclusion of a 10-element seed:*

$$S = \{0, 1, 2, 3, 5, 7, 9, 12, 15, 35\}.$$

Proof. • C1 forces $0 \in \mathcal{C}$.

- C4 forces $35 \in \mathcal{C}$ (as maximum).
- C6(c) forces $\{1, 2, 3, 5, 7, 9, 15\} \subseteq \mathcal{C}$.
- C2 and C3 force $12 \in \mathcal{C}$ (since $|\mathcal{C}| = 12$ must be a member).

These nine elements are distinct. Together with 35, we have $|S| = 10$. □

Lemma 4.2 (Completion count). *Any set $\mathcal{C}$ satisfying C1–C4 and C6(c) must be a 12-element subset of $\{0, 1, \dots, 35\}$ containing the seed S . Exactly two elements must be added to S from the pool $P = \{0, \dots, 35\} \setminus S$, giving $\binom{26}{2} = 325$ candidate completions.*

Proof. C3 fixes $|\mathcal{C}| = 12$. C4 fixes $\max(\mathcal{C}) = 35$. By C1, $\min(\mathcal{C}) = 0$. Thus $\mathcal{C} \subseteq \{0, \dots, 35\}$. Since $|S| = 10$ and $|\mathcal{C}| = 12$, exactly two elements from P must be added. The pool has $36 - 10 = 26$ elements. □

Theorem 4.3 (Uniqueness). *The Coralia Sequence $\mathcal{C} = \{0, 1, 2, 3, 5, 7, 9, 12, 15, 23, 30, 35\}$ is the unique 12-element subset of $\{0, \dots, 35\}$ satisfying axioms C1–C9.*

Proof. By Lemmas 4.1 and 4.2, any solution must be a completion of the seed S by exactly two elements from the 26-element pool P .

We verify uniqueness by exhaustive enumeration. For each of the $\binom{26}{2} = 325$ candidate pairs $\{a, b\} \subset P$, we form $\mathcal{C}' = S \cup \{a, b\}$, compute its ordered enumeration and gap sequence, and check axioms C5–C9:

- C5: $\sum g_i = 35$,
- C6(a): $g_0, \dots, g_7 \in \{1, 2, 3\}$,
- C6(b): $g_0 \leq g_1 \leq \dots \leq g_7$,
- C7: $g_8 > g_9 > g_{10}$,
- C8: $g_8, g_9, g_{10} \in \text{Fib} \cup \text{Luc}$,
- C9: $g_8, g_9, g_{10} \geq 5$.

Exactly one completion passes all checks: $\{23, 30\}$, yielding $\mathcal{C} = \{0, 1, 2, 3, 5, 7, 9, 12, 15, 23, 30, 35\}$. Since the candidate space is finite (325 cases), exhaustive verification constitutes a complete proof.

Verification. The enumeration is implemented in Python and available at:

<https://github.com/coralio-io/coralia-sequence>

Running `verify_uniqueness.py` confirms: Solutions: 1. □

5 Terminal Triple Uniqueness

Theorem 5.1 (Terminal triple uniqueness). *Let $\text{FL} = (\text{Fib} \cup \text{Luc}) \cap \{1, 2, \dots, 20\}$. The triple $(8, 7, 5)$ is the unique strictly descending triple (a, b, c) with $a, b, c \in \text{FL}$, $a + b + c = 20$, and $c \geq 5$.*

Proof. We have $\text{FL} = \{1, 2, 3, 4, 5, 7, 8, 11, 13, 18\}$.

Since $a > b > c \geq 1$ and $a + b + c = 20$, we have $a \geq 8$. The candidates $a \in \text{FL} \cap \{8, 11, 13, 18\}$ yield:

a	Descending triples (a, b, c) with $b + c = 20 - a$
18	None (would need $b + c = 2$ with $b > c \geq 1$)
13	$(13, 5, 2)$, $(13, 4, 3)$
11	$(11, 8, 1)$, $(11, 7, 2)$, $(11, 5, 4)$
8	$(8, 7, 5)$

All six triples are valid under the first three constraints. Since $a > b > c$, we have $\min(a, b, c) = c$. Applying $c \geq 5$, only $(8, 7, 5)$ survives.

The Fibonacci–Lucas identities are: $8 = F_6$, $7 = L_4$, $5 = F_5$. □

6 Structural Properties

Proposition 6.1 (Closure properties). *The Coralia Sequence satisfies:*

- (i) **Cardinality containment:** $|\mathcal{C}| = 12 \in \mathcal{C}$.
- (ii) **Gap-sum closure:** $\sum \text{gaps}(\mathcal{C}) = \max(\mathcal{C}) = 35$.
- (iii) **Cumulative generation:** *The cumulative sums of $\text{gaps}(\mathcal{C})$ are $\mathcal{C} \setminus \{0\}$.*

Proof. Direct verification: $\text{gaps}(\mathcal{C}) = (1, 1, 1, 2, 2, 2, 3, 3, 8, 7, 5)$.

- (i) $12 \in \mathcal{C}$ by inspection.
- (ii) $1 + 1 + 1 + 2 + 2 + 2 + 3 + 3 + 8 + 7 + 5 = 35 = \max(\mathcal{C})$.
- (iii) Cumulative sums: $1, 2, 3, 5, 7, 9, 12, 15, 23, 30, 35 = \mathcal{C} \setminus \{0\}$. □

7 Open Questions

1. Does a natural extension of $\mathcal{C}$ exist beyond 35 preserving the axiom structure?
2. Is there a closed-form characterization of the two free elements $(23, 30)$?
3. Do analogous sets exist for other terminal triples from $\text{Fib} \cup \text{Luc}$?

References

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